Nuclear Physics and Spectroscopic Techniques

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Nuclear Physics and Spectroscopic Techniques (A29521)

Short Title:	Nuclear Physics and Spec Tech
Faculty	Science and Computing
Department	Science
Cluster	Mathematics and Physics
Author	CORAIFEARTAIGH
Credits	5
Level:	Advanced

Description of Module / Aims

This module covers two distinct topics of fundamental importance in modern physics: the physics of the atomic nucleus and the principles underlying a range of advanced spectroscopic techniques.

Programmes

PHYI-0041	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	4 / 7 / M

Indicative Content

- The Nucleus: review of the nucleus and radiation; nuclear instability and energy levels
- Nuclear collisions and reactions: nuclear fission and the thermal fission reactor
- Nuclear models including Liquid Drop Model and SEMF, Shell Model
- Interaction of radiation with matter: radiation detection techniques including high-resolution gamma ray spectroscopy
- Generation and Nature of X-rays; X-ray Photoelectron Spectroscopy (XPS) and Auger Electron Spectroscopy (AES)
- Scanning Electron Microscopy (SEM); EDX and WDX spectroscopy
- Ion beam techniques and Secondary Ion Mass spectroscopy (SIMS)

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Appraise the basic properties of the nucleus, the factors affecting the stability of the nucleus and perform routine calculations using decay laws.
2. Appraise the concept of a nuclear cross section via nuclear collision reactions and perform calculations.
3. Construct a model of the nucleus using the Liquid Drop Model and the Shell Model and apply the SEMF to determine nuclear binding energy.
4. Appraise the principles of radiation detection and measurement.
5. Appraise the principles of a number of advanced spectroscopic techniques.
6. Test the knowledge gained in class using experiments involving gamma-ray spectroscopy, SEM and mass spectroscopy.

Learning and Teaching Methods

- Lectures.
- Tutorials.
- Practicals.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,4,5
Continuous Assessment	30%	
Practical	30%	6

Assessment Criteria

<40%: Unable to demonstrate a basic understanding of course content.
40%-49%: Able to (i) describe some basic nuclear properties and structure, (ii) give basic description of simple models of the nucleus and (iii) give a basic description of the standard model of particle physics.
50%-59%: Able to (i) adequately describe basic nuclear properties and structure, and perform nuclear calculations and (ii) give adequate descriptions of simple models of the nucleus and of the standard model of particle physics.
60%-69%: Able to apply given problem-solving techniques to new similar problems.
70%-100%: All the previous to an excellent level.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Supplementary Material

- Brandon, D. and W. Kaplan. _Microstructural Characterization of Materials_. 2nd ed. New York: Wiley, 2008.
- Hollas, J.M. _Modern Spectroscopy_. 4th ed. New York: Wiley, 2003.
- Krane, K. _Introductory Nuclear Physics_. New York: Wiley, 1988.
- Lilley, J. _Nuclear Physics: Principles and Applications_. London: Wiley, 2001.
- Milner, B. _Nuclear and Particle Physics: An Introduction_. 2nd ed. New York: Wiley Blackwell, 2009.
- Runyan, W.R. _Semiconductor Measurements and Instrumentation_. New York: McGraw-Hill, 1998.
- Williams, S.C. _Nuclear and Particle Physics_. Oxford: Clarendon Press, 1991.

Requested Resources

- SCIENCE LAB: Physics